

PH 702 Introduction to Biostatistics

Part 0: Syllabus Review and Beyond

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Contact

- Instructor: Zhiyang Zhou
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 - Homepage: zhiyanggeezhou.github.io
- Meeting Times
 - Section 401 (lecture): Mon 5:30–7:20 pm @Zilber Triplex
 - Section 802 (lab): Thur 2:30–4:20 pm @Zilber 240
 - Section 801 (lab): Thur 7:30–9:20 pm @Zilber 240
 - Section 202 (lecture): asynchronous
 - Section 901 (lab): asynchronous
- Office Hours
 - Mon 4:00–5:00 pm @Zilber 376

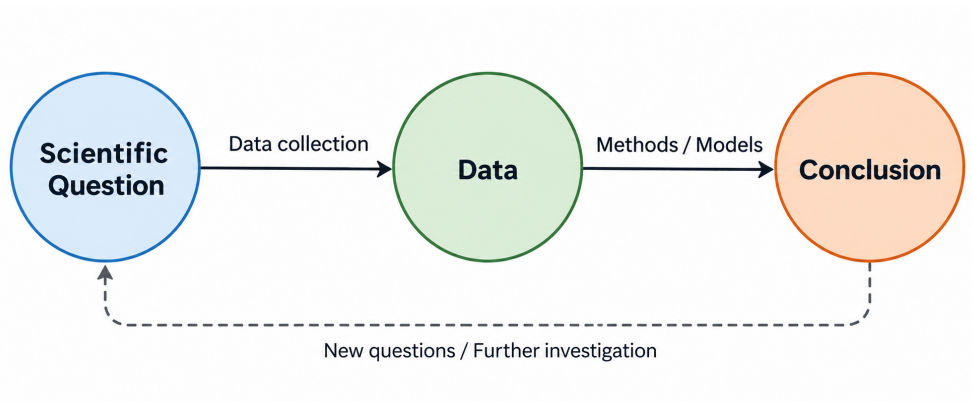
Grading

- Assignments (60%)
 - Submitting via Canvas
 - Attaching both outputs and source codes (if applicable)
 - Including necessary interpretation
 - Organized in a CLEAR and READABLE way
 - Accepting NO late submission
- Exams (40%)
 - Midterm (20%; Oct. 26)
 - Final (20%; TBD)

Materials

- Lecture notes and beyond
 - Posted on Canvas and zhiyanggeezhou.github.io
 - Subject to update **without prior notice**
- Recommended but not required
 - D. Moore, G. McCabe, & B. Craig. (2017). *Introduction to the Practice of Statistics*, 9th Ed. New York: W. H. Freeman & Co.
 - L. Delwiche & S. Slaughter. (2012). *The Little SAS Book: A Primer*, 5th Ed. Cary, NC: SAS Institute Inc.

Big picture of (bio)statistics



AI can support (bio)statistics at each stage

- Scientific question: Synthesize literature, identify knowledge gaps, refine research questions, generate hypotheses.
- Data collection: Design surveys/questionnaires, identify variables, support sampling and recruitment, extract information from EHRs/text/images, automate measurement and data capture.
- Data: Clean and code data, integrate data sources, conduct quality checks, visualize data, support exploratory analysis.
- Methods & Models: Suggest candidate methods, generate code, fit models, check assumptions, perform diagnostics and sensitivity analyses.
- Conclusion: Interpret results, summarize evidence, generate tables/figures, and communicate findings.

Why learn the biostatistics in this AI era?

AI can calculate faster, write code, fit models, and generate interpretations. But we still need statistical thinking to ask the right questions and judge the answers.

- Ask the right question: Turn a scientific question into something that can be answered with data.
- Understand the data: Know what was measured, how it was collected, and what biases or limitations it contains.
- Choose and evaluate methods: Understand what a method/model assumes and whether it is appropriate.
- Critically evaluate AI output: AI can produce convincing calculations, code, and explanations that are still statistically wrong.
- Make responsible conclusions: Ultimately, researchers must decide what the evidence supports.